Fir Filter Design Space Exploration

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Design Trade-offs – Area vs. Performance

FIR Filter Design

- FIR Filter Design
  - Given an impulse input, the filter output goes to zero in a finite number of clocks because there is no feedback of the output to the input
  - Filter: Manipulate the frequency response
  - Examples: low-pass, high-pass, band pass, notch, arbitrary
  - Equation:

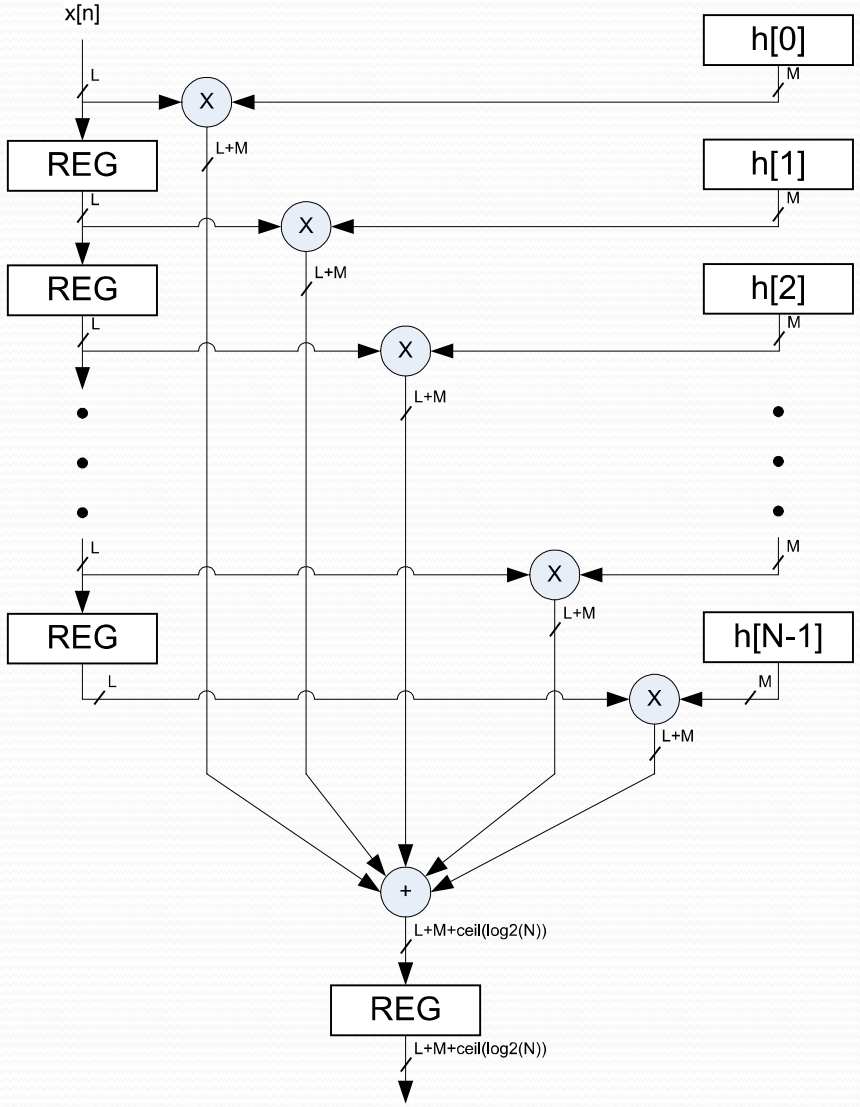
$$y[n] = \sum_{m=0}^{N-1} x[n-m] \cdot h[m]$$

y : output
x : input
h : filter coefficients a.k.a filter taps
N : number of coefficients

Design Trade-offs – Area vs. Performance

FIR Filter Design – Parallel Implementation

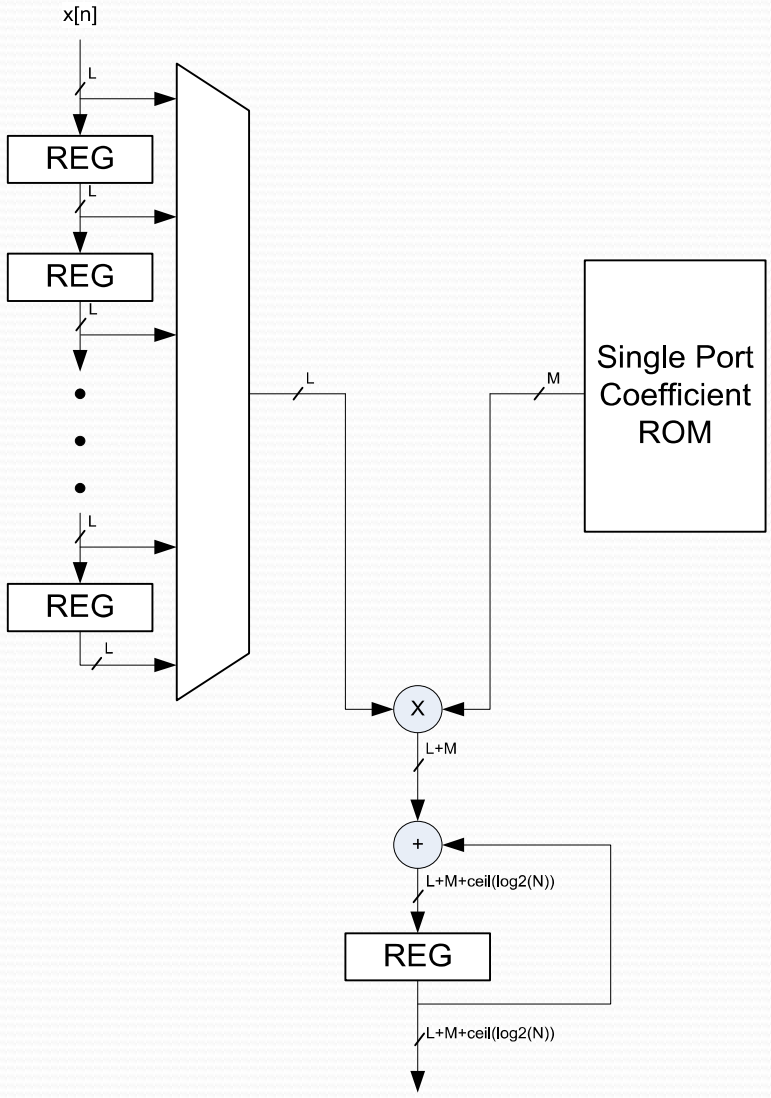
- Output every clock
- One multiplier per tap
- N-input adder
- x[n]: shift register
- h[n]: stored constants
- Note the bit growth on the output register/adder



Design Trade-offs – Area vs. Performance

FIR Filter Design – Serial Implementation

- Output every N clocks
- One multiplier
- Two input Adder
- Multiply-Accumulate (MACC)
- Note the bit growth on the output register/adder



Design Trade-offs – Area vs. Performance

FIR Filter Design – Compromised Implementation

- K multipliers
- Output every N/2 clocks
- 2+1 input Adder
- Multiply-Accumulate (MAC)
- Note the bit growth on the output register/adder

